

# Tahsin Tajwar Tanni

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[Portfolio](#) | [LinkedIn](#) | [Github](#) | [Google Scholar](#)

## Profile

I'm a Computer Science and Engineering graduate from BRAC University with experience spanning AI automation, machine learning research, and full-stack web development. My interests lie in artificial intelligence, machine learning, agentic systems, full-stack development, blockchain, and building impactful technologies that solve real-world problems.

## Skills

- **Programming Languages:** Python, JavaScript, TypeScript, C, Solidity
- **Databases:** MySQL, MongoDB, SQLite, Prisma ORM
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, NumPy, pandas, scikit-learn, PyTorch, Matplotlib, Seaborn.
- **AI & Automation:** OpenAI API, Anthropic API, n8n, RAG, Prompt Engineering
- **CSS Frameworks:** Tailwind CSS
- **Tools & Platforms:** Git, GitHub, Postman, Hugging Face Spaces, Docker.

## Education

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|-----------------------------------------------------------------------------------------------------|-----------------------|
| <b>BRAC University</b> , BSc in Computer Science and Engineering                                    | 2022 – February, 2026 |
| • GPA: 3.60 / 4.00                                                                                  |                       |
| <b>Govt. Haji Muhammad Mohsin College</b> , Higher Secondary Certificate (HSC)                      | 2018 – 2020           |
| • GPA: 5.00 / 5.00                                                                                  |                       |
| <b>Bangladesh Mohila Samity Girls' High School and College</b> , Secondary School Certificate (SSC) | 2004 – 2018           |
| • GPA: 5.00 / 5.00                                                                                  |                       |

## Experience

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|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>AI Automation Trainee</b> , 180 RE, Toronto, Canada                                                                                                  | May 2025 – Present        |
| • Built LLM-powered workflows using OpenAI and Anthropic APIs, automating multi-step business processes via n8n and third-party service integrations.   |                           |
| • Developed and deployed AI-powered chatbots, integrating them into websites to automate user interactions and support.                                 |                           |
| • Applied prompt engineering techniques to optimise LLM outputs for accuracy and safety across use cases.                                               |                           |
| <b>Python Tutor</b> , Zentorra                                                                                                                          | February 2025 – May 2025  |
| • Tutored high school students in Python programming, covering fundamentals through advanced topics, and assisted with coding assignments and projects. |                           |
| <b>Junior Web Developer (Part-time)</b> , Weabers                                                                                                       | February 2023 – July 2024 |
| • Developed and maintained responsive web applications, building reusable UI components and integrating APIs with backend teams.                        |                           |
| • Debugged issues, refactored code, and deployed feature updates across production codebases.                                                           |                           |
| <b>Junior Executive</b> , IT Department, Robotics Club of BRAC University                                                                               | June 2022 – August 2023   |
| • Collaborated with team members to manage internal digital platforms and technical resources.                                                          |                           |

## Projects

### ProfMail – AI-Powered Academic Cold Email Generator

- An AI-driven web application that automates personalized cold email generation for graduate school applicants, featuring real-time professor lab scraping and research overlap analysis between student thesis and faculty publications.
- Includes customizable tone and length controls with an interactive editing workspace and local email history.
- Tools Used: React, TypeScript, Vite 6, Tailwind CSS

### AI Professor Finder

- An automated n8n workflow that discovers and evaluates potential MSc/PhD supervisors using Google Scholar and Gemini AI.
- Reads professor information from Google Sheets, retrieves recent publications via SerpAPI, compares research against a student's thesis, and produces AI-generated compatibility scores and funding recommendations.
- Tools Used: n8n, Google Scholar, SerpAPI, Gemini AI, Google Sheets

### Realtime Collaborative Playlist

- Built a real-time collaborative music playlist application with Server-Sent Events (SSE) for live synchronization, drag-and-drop reordering using fractional indexing, a voting system, and now-playing simulation.
- Tools Used: Next.js 16, React 19, TypeScript, Prisma ORM, SQLite, Tailwind CSS 4, Framer Motion

### Thesis-Management-System-ThesisFlow

- This project Thesis Management System is a digital platform that streamlines thesis submission, tracking, and supervision. It enhances communication, organization, and efficiency in managing academic research projects
- Tools Used: MongoDB, Express.js, React, Node.js, HTML, CSS

### Stock Forecasting Analysis Toolkit

- Developed an interactive Hugging Face Space demonstrating data synthesis using generative AI techniques for job-related analytics.
- Enabled dynamic user interaction for synthetic data generation and visualization through a web-based interface.
- Tools Used: pandas, numpy, matplotlib, statsmodels, scikit-learn, prophet, tensorflow, gradio, Hugging Face

### vae-dec-clustering-uncertainty

- This project is the implementation and analysis of a Variational Autoencoder (VAE) combined with Deep Embedded Clustering (DEC) for unsupervised image clustering and also explores uncertainty quantification (via entropy and latent variance) to evaluate clustering correctness and model confidence.
- Tools Used: python, pytorch, pandas, NumPy, scikit-learn, matplotlib, seaborn

### Software-Course-Management-System

- This project is a web-based platform for managing university courses, allowing administrators to create courses, assign instructors, enroll students, and track academic progress efficiently.
- Tools Used: HTML, CSS, Node.js, React, MongoDB

## Research Experience

### Undergraduate Researcher

January 2025 – January 2026

BRAC University, Department of Computer Science and Engineering

- Investigated post-deployment vulnerabilities in pretrained language models – weight-noise injection and low-precision quantization attacks that degrade stability without altering architecture.
- Designed **Anchor-Guided Repair**, a fine-tuning defense jointly optimizing language modeling loss and anchor regularization to constrain parameter drift from a clean baseline.
- Validated across multiple LLM architectures and attack types, consistently recovering performance close to the clean baseline.

**Publication:** Rohan, Khan, Tanni, Fardin, & Bushra (2026). Anchor-Guided Repair: A Defense Mechanism for Enhancing Stability of Compromised Pretrained Language Models. *IndabaX Nigeria 2026*, PMLR 319, 368–381.  

### Phantom Guard: Fine-tuned Multi-Agent System for Offline Slopsquatting Detection

Ongoing

Role: Researcher

- Designing a dual-LLM generator-reviewer pipeline with registry/API fallback validation to detect hallucinated, slopsquatted package recommendations.
- Exploring knowledge distillation for lightweight, low-latency self-verifying models.

## Achievement

- Achieved **2nd Runner-up** at the **CSE471 Project Demonstration** with *ThesisFlow*, a comprehensive **Thesis Management System** designed to centralize and streamline the entire thesis lifecycle at BRAC University, featuring role-based dashboards, progress tracking, structured feedback, and AI-powered academic guidance.
- Successfully completed an **Intermediate Web Development** course during the Residential Semester at BRAC University, gaining practical skills in designing responsive websites.
- Completed the **Intro to Machine Learning** course on Kaggle, gaining foundational knowledge in data science and machine learning with practical applications to real-world problems.
- Completed **AI Prompt Engineer™ (AI CERTs™)** via AgentX Bangladesh (Netcom Learning & Microsoft). Skills: advanced prompt design (zero / few shots, chain of thought), PFSET / PAIPS, IPS, XML steering, agent workflows, and ethical AI use.